Combinational
Circuits

SAMARTH

CUET PG 2027

Lecture - 12 | Computer Science

Unit 03 : Digital Fundamental
Topic : Combinational Circuits

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Recap *of previous lecture*

1. Kmap Simplification: Part 2



Topics *to be covered*

1. Combinational Circuits



Topic : Combinational Circuits

Topic : Combinational Circuits

Input → o/p.

- A combinational circuit is a digital circuit whose output depends only on the present input. Combinational circuits can be made using various logic gates, such as AND gates, OR gates, and NOT gates.

There are the following characteristics of the combinational logic circuit:

- At any instant of time, the output of the combinational circuits depends only on the present input terminals.

Ex Gates

A	B	A·B
0	0	0
0	1	0
1	0	0
1	1	1

Ex Calculator

Ex ALU

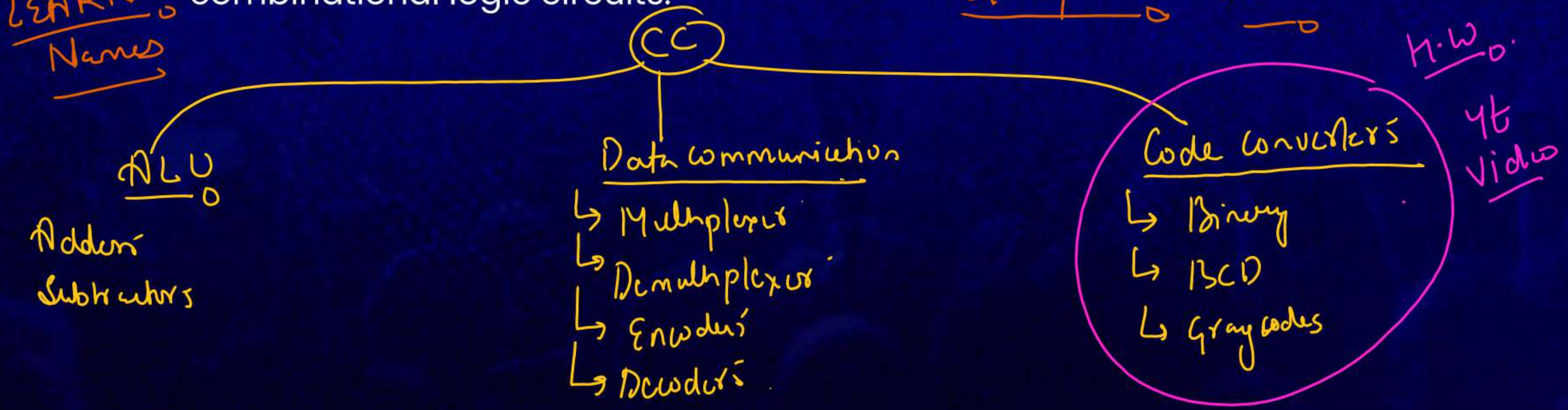
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No memory

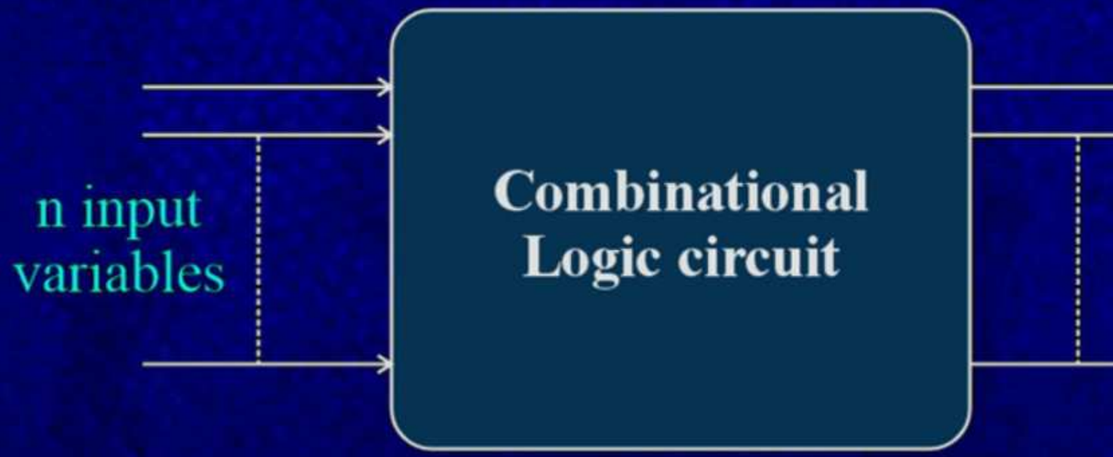
- The combinational circuit doesn't have any backup or previous memory. The present state of the circuit is not affected by the previous state of the input.
- The n number of inputs and m number of outputs are possible in combinational logic circuits.

LEARN Names

Q Expected Direct



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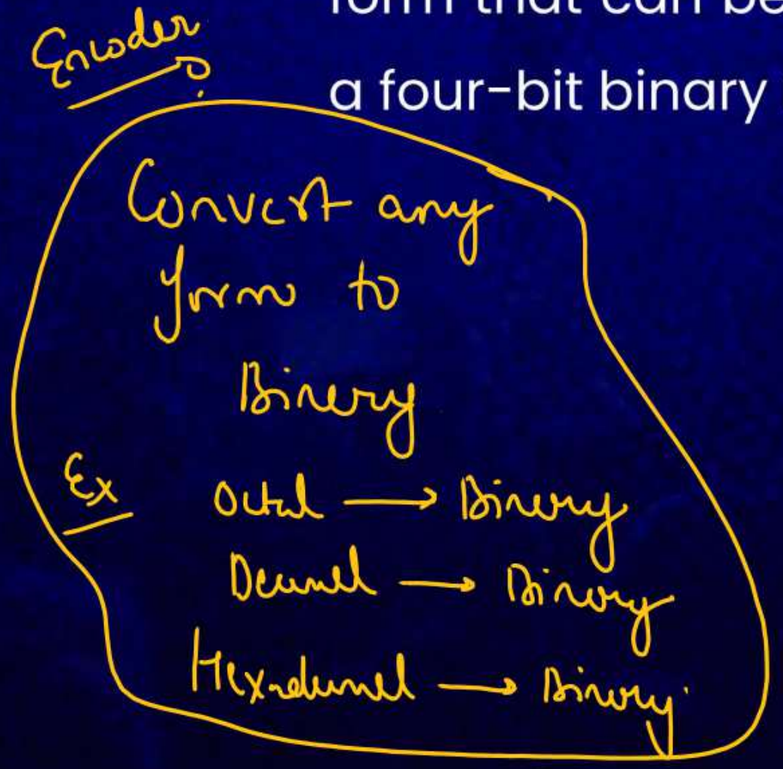
Block diagram of a combinational circuit

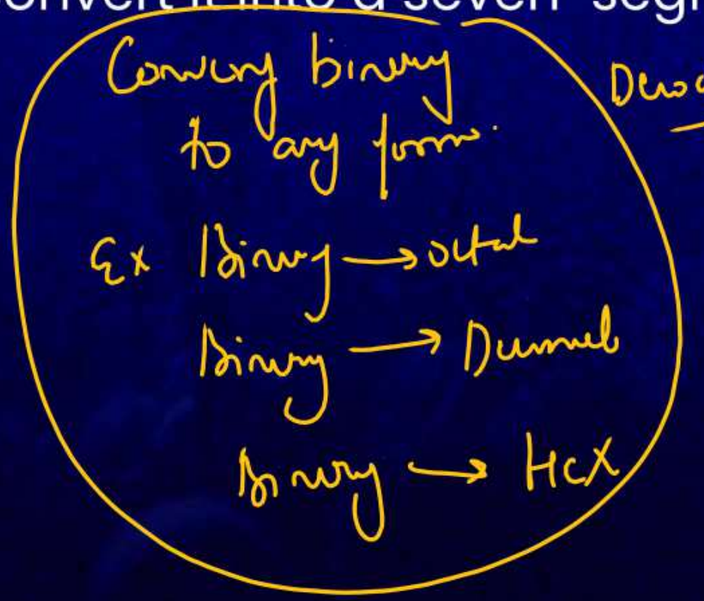
- The 'n' input variable comes from the external source while the 'm' output variable goes to the external destination. In many applications, the source or destinations are storage registers.

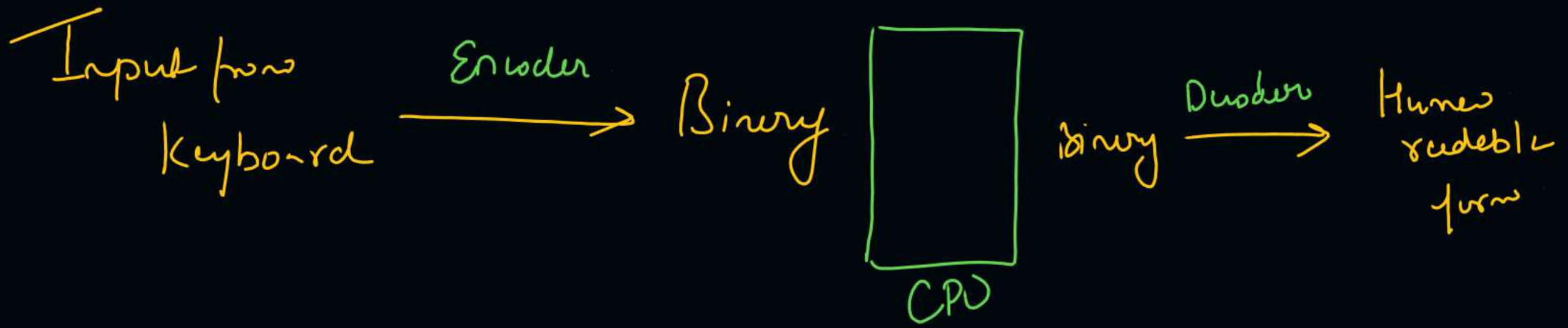
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There are three main types of combinational circuits: decoders, encoders, and multiplexers.

1. Decoders are combinational circuits that convert binary code into a form that can be read by a human. For example, a decoder might take a four-bit binary number and convert it into a seven-segment display.





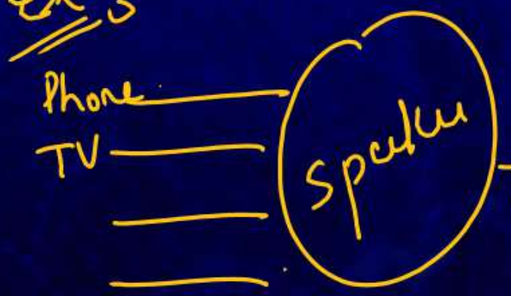

Input from keyboard
Encoder
Binary
CPU
Binary
Decoder
Human readable form

2. Encoders are combinational circuits that convert information from one form to another. For example, an encoder might take a seven-segment display and convert it into a four-bit binary number.
3. Multiplexers are combinational circuits that select one of several input signals and send it to the output. For example, a multiplexer might take four input signals and send only one of them to the output.

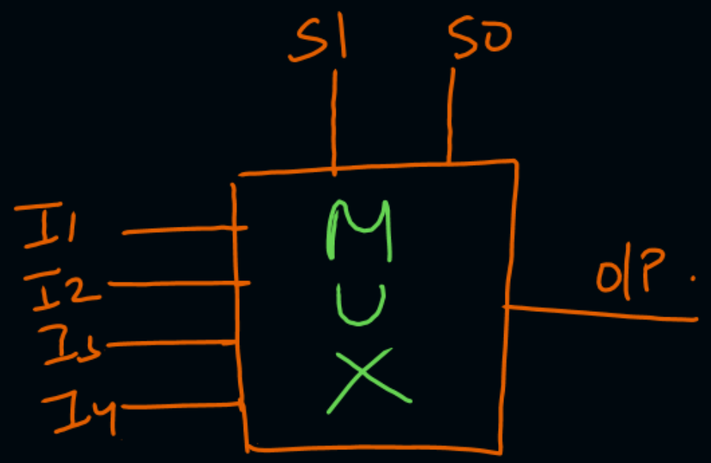
Also called as Data selector.



Ex



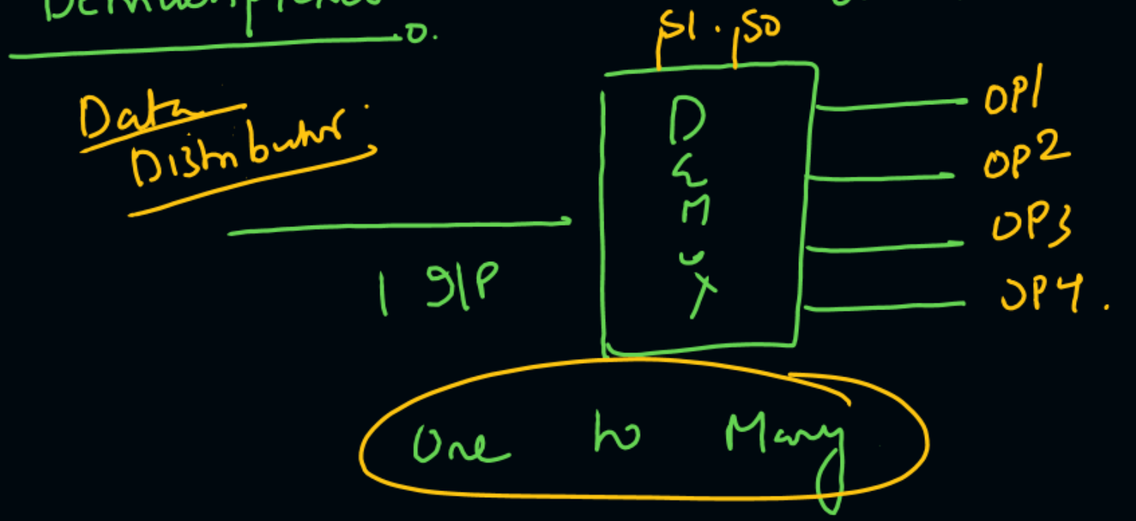
Ex DJ



S1	S0	O/P
0	0	I1
0	1	I2
1	0	I3
1	1	I4

Demultiplexer = 1 to 2^n.

Data Distributor.



S1	S0	OP1	OP2	OP3	OP4
0	0	✓	X	X	X
0	1	X	✓	X	X
1	0	X	X	✓	X
1	1	X	X	X	✓

Imp. Adder → Half adder – To add 2 bits
Full Adder – To add 3 bits.

HALF ADDER VS FULL ADDER:

- There is a primary difference between half adder and full adder. Half adder only adds the current inputs as 1-bit numbers and does not focus on the previous inputs. On the other hand, Full Adder can easily carry the current inputs as well as the output from the previous additions.

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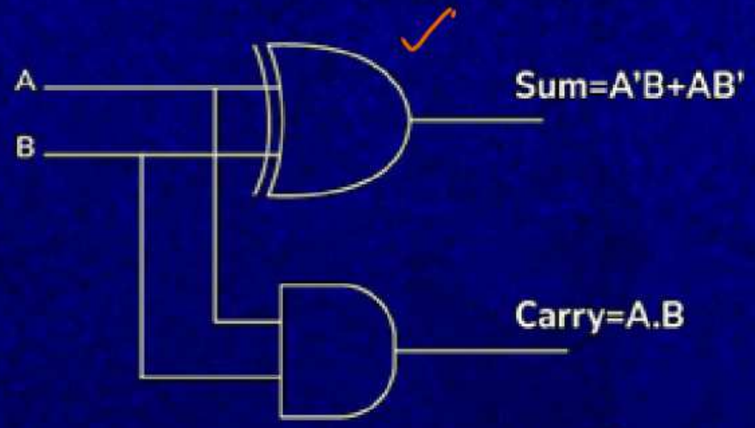
Logical Expression of Half Adder

Sum (S) = $A \oplus B$

Carry (C) = $A \cdot B$

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Half adder

Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



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Logical Expression of Full Adder

CARRY-OUT = AB + BCin + ACin

SUM = (A ⊕ B) ⊕ Cin

X	y	C_i	S	C_o
1	0	0	1	0

① let
$X=1, y=0, C_i=0$

Find sum & carry out in full adder.

① $S=1, C_o=0$

② $S=0, C_o=1$

③ $S=0, C_o=0$

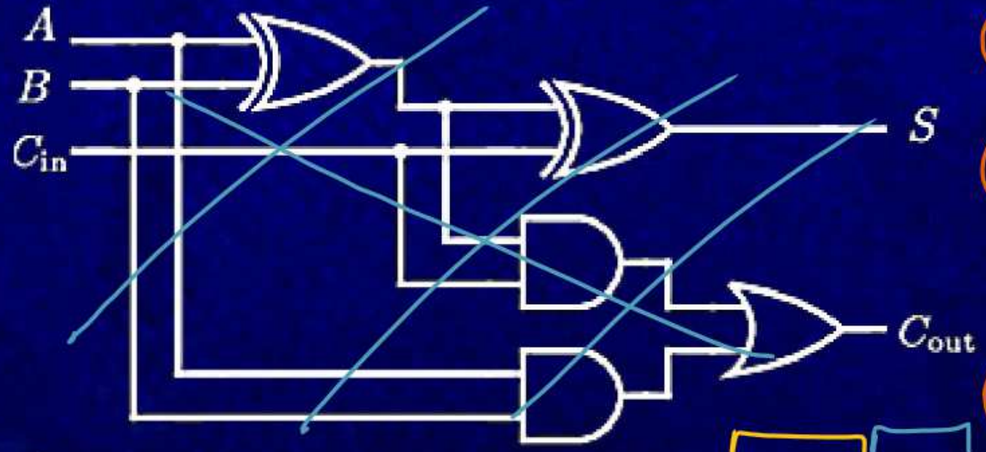
④ $S=1, C_o=1$

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Ex = 1101 + 1011 = 11000

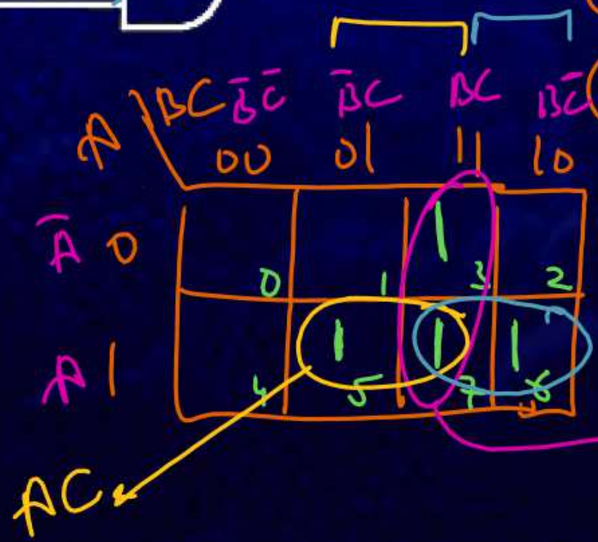
(1) All 3 input 0 — Sum 0 carry 0
(2) Any 1 input 1 — Sum 1 carry 0
(3) Any 2 input 1 — Sum 0 carry 1
(4) All input 1 — Sum 1 carry 1



	Inputs			Outputs		
	A	B	C_{in}	S	C_{out}	
(1)	0	0	0	0	0	0
(2)	0	0	1	1	0	1
	0	1	0	1	0	2
(3)	0	1	1	0	1	3
(2)	1	0	0	1	0	4
(3)	1	0	1	0	1	5
	1	1	0	0	1	6
(4)	1	1	1	1	1	7

$$Sum = A \oplus B \oplus C$$

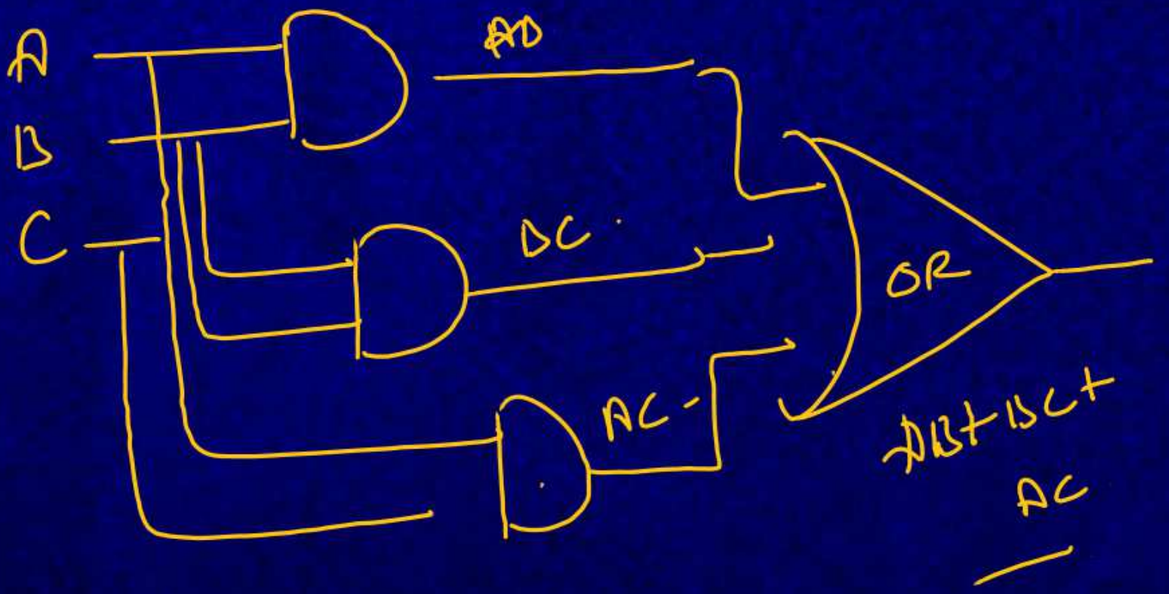
$$Carry = AB + AC + BC$$



$AB + AC + BC$

2 Mins Summary

- CC
- Adder
  - HA
  - FA
- MUX
- DEMUX
- Encoder
- Decoder



Homework from SAMARTH CUET PG 2027 Module

Expected Type of Q

Q Find True.

S1: o/p of CC depend only on current input. ✓ ∴ No memory

S2: o/p of CC depend on both current input & previous o/p ✗

(1) S1 (2) S2 (3) Both (4) None

Q Which of the foll is/are an example of CC.

(1) Flip Flop — SC.

(2) ✓ Encoder

(3) ✓ Adder

(4) Register — SC.

(5) ✓ Multiplexer.

(A) 1, 3, 5 (B) ✗ 2, 3, 5 (C) 2, 3, 4 (D) 1, 2, 5

THANK YOU